

Xudong Zhu

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My research examines how large language models represent concepts and how these representations evolve during generation and training. I focus on their linear structure, how training reshapes it, and its role in understanding and steering model behavior.

RESEARCH INTERESTS

Representation Learning · Interpretability · Training Dynamics
Representation & Behavior Steering · Agent Behavior Control

EDUCATION

The Ohio State University Ph.D. in Computer Science and Engineering, advised by Prof. Zhihui Zhu	2024 – Present Columbus, OH
University of Electronic Science and Technology of China B.S. in Computer Science · GPA 3.98 / 4.00	2020 – 2024 Chengdu, China

PUBLICATIONS

- AbsTopK: Rethinking Sparse Autoencoders for Bidirectional Features**
Xudong Zhu, Mohammad Mahdi Khalili, Zhihui Zhu
[ICLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)
- From Emergence to Control: Probing and Modulating Self-Reflection in Language Models**
Xudong Zhu, Jiachen Jiang, Mohammad Mahdi Khalili, Zhihui Zhu
[TMLR 2026](#) · [Paper](#) · [Code](#) · [Project Page](#)
- FCDS: Fusing Constituency and Dependency Syntax into Document-Level Relation Extraction**
Xudong Zhu, Zhao Kang, Bei Hui
[LREC-COLING 2024](#) · [Paper](#) · [Code](#) · [Project Page](#)
- Alleviating Subgraph-Induced Oversmoothing in Link Prediction via Coarse Graining**
Xudong Zhu, Dong Hao, Ziqin Gao, Liming Pan
[Neurocomputing, 2025](#) · [Paper](#) · [Code](#) · [Project Page](#)

HONORS & RESEARCH SUPPORT

MMLS 2026 Travel Award , Midwest Machine Learning Symposium	2026
NAIRR Pilot Project NAIRR260106 (contributor), National AI Research Resource Pilot Geometry and Training Dynamics of Representations in Large Language Models	2026

OPEN SOURCE

LLMUsage · Local-first LLM subscription usage monitor for Hyprrland (OpenAI, GLM, Grok, Claude)	TypeScript
dotfiles · fish, Neovim, and OpenCode configuration managed with GNU Stow	Lua